
HACKATHON SUBMISSION | VOICE AI & EDTECH

NOVAFUSE

Voice-First AI Study & STAR Interview Agent

Transforming passive study habits into high-retention, real-time spoken dialogue with adaptive evaluation.

KEY FEATURES

- Socratic Active Recall Revision Mode

Live App: <https://www.novafuse.site/dashboard>

Presenter: Sahil Rawat | Powered by AssemblyAI & WebSockets

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THE PROBLEM

Silent Reading Creates an Illusion of Competence

Traditional learning platforms test recognition, not verbal recall under pressure.

Passive Recognition

Students re-read notes and click multiple-choice buttons, which tests visual cues rather than deep mental synthesis.

Result: Knowledge collapses during spoken exams.

Verbal Articulation Gap

Spoken explanation engages separate neurological circuits: pacing, real-time structuring, and technical vocabulary.

Result: High anxiety & hesitation in real interviews.

Coaching Bottlenecks

Human mock interviewers and viva tutors cost \$100+/hr, require scheduling across zones, and induce peer fear.

Result: Practice happens too late or not at all.

THE SOLUTION

NovaFuse: Your Autonomous AI Spoken Dialogue Partner

A voice-first agent running full-duplex spoken dialogues with adaptive memory.

1. Active Recall Revision Mode

- Socratic inquiry: Quizzes concepts, probes deeper on vague answers
- Real-time pacing: Adapts difficulty dynamically on streaks
- Instant remediation: Breaks down complex topics into analogies
- Function calling: Evaluates answer quality in real-time

Target: University students, viva prep, medical licensing

2. STAR Mock Interview Studio

- Strict STAR enforcement: Situation, Task, Action, Result
- Technical role calibration: SWE, System Design, Product Leads
- Probing follow-ups: Pushes on metrics, architecture trade-offs
- Comprehensive rubric: Delivers structured hire recommendation

Target: Software engineers, job seekers, behavioral loops

Sub-Second Latency Full-Duplex Audio Pipeline

Built with browser AudioWorklets, Node.js WebSocket relay, and AssemblyAI.

1. AudioWorklet

Captures microphone audio off main thread.

Resamples to 24kHz PCM16 mono audio with zero jitter.

Web Audio API

2. WebSocket Relay

Node.js/ws proxy protects secret keys.

Relays binary PCM and JSON events with 30s reconnect.

Node.js + Express

3. AssemblyAI Agent

Unified WebSocket: STT + LLM + TTS.

Neural turn detection and barge-in voice interruption support.

AssemblyAI Voice API

4. Data Layer

PostgreSQL & Prisma ORM.

Self-healing DDL auto-migration on server boot.

Prisma + Postgres

KEY INNOVATION

Continuous Memory: The Weak-Topic Injection Loop

Autonomous spaced repetition powered by real-time function calling.

1. In-Call Evaluation

Agent triggers log_answer_quality while talking. Stores status (strong/partial/weak) & clinical misconception notes.

2. Spaced Extraction

When the student starts a new session, historical weak topics are automatically queried and ranked by urgency.

3. Adaptive Prompt

System prompt instructs agent to naturally re-examine weak points without making the learner feel discouraged.

TRANSPARENT UNIT ECONOMICS

Calculates exact session duration and API cost at \$0.075/min.
Full transparency for both individual learners and educational institutions.

RESULTS & ROADMAP

Shipped, Live & Verified in Production

Deployed on Render with 40/40 passing integration tests.

What We Accomplished

- Real-time full-duplex spoken conversation in browser
- Barge-in interruptions & neural semantic turn-taking
- Dual Socratic Revision + STAR Interview evaluation
- Self-healing PostgreSQL schema bootstrapping on boot

Live App: <https://www.novafuse.site/dashboard>

What is Next for NovaFuse

- Multi-agent panel interviews (e.g. Hiring Manager + Tech Lead)
- Computer vision analysis for eye contact and body poise
- Native Anki & Notion sync for weak-topic flashcards
- Multi-language oral proficiency exam simulation

NovaFuse: Master Knowledge Through Spoken Voice